

ADITHYA BHAT

Visa Research, Visa Inc., CA

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SUMMARY

Distributed systems engineer with a Ph.D. and 8+ years designing, building, and evaluating fault-tolerant, high-performance distributed systems. Deep expertise in consensus protocols (Paxos, BFT, state machine replication), replication, and multi-threaded concurrent systems. Author of 13 peer-reviewed publications at CCS, IEEE S&P, NDSS, FC, and Middleware; 3 US patents. Proficient in C/C++, Rust, Go, and Python, with hands-on performance engineering across networking, storage, and consensus hot paths.

TECHNICAL SKILLS

Languages	C/C++, Rust, Go, Python
Distributed Systems	Consensus (Paxos, BFT, CFT), State Machine Replication, Replication, Sharding, High-Availability, Fault-tolerance
Performance & Concurrency	Multi-threading, async I/O, lock-free data structures, Performance analysis, benchmarking, profiling, hot-path optimization
Systems & Infra	Linux, TCP/TLS networking, RocksDB, gRPC, AWS, Docker, Git

WORK EXPERIENCE

Visa Research, Foster City (Aug 2023 - Present)
Staff Research Scientist

- Led research on fault-tolerant distributed systems for payments infrastructure, spanning consensus, storage, primary-backup replication acceleration, and empirical scaling analysis.
- Invented hint-based replication acceleration: filed US patent on *FastSync* (WO2024026321A1, Belady-optimal cache replacement via clairvoyant hints from already-synchronized nodes) and generalized it as *Ira* (2026 tech report), a primary-backup replay framework delivering $5.2 \times - 23.6 \times$ backup speedup at $\approx 11\%$ primary overhead across 24M Ethereum mainnet transactions (100k blocks).
- Authored *Adhara* (2026, under submission), a year-scale empirical study of blockchain execution scaling over **42M** Ethereum transactions / 208K blocks: demonstrated that state sharding caps at $1.37 \times$ even under ideal offline partitioning due to hub-contract skew, while intra-block *parallel execution via per-block conflict DAGs* achieves a $13.68 \times$ analytical parallelism bound with zero speculation and zero aborts.
- Designed and implemented *FPS* (Flexible Payment Systems tech report, 2025): a BFT payment protocol with 15k LOC Rust implementation across 16 crates on a self-designed networking/crypto/storage substrate, with reproducible multi-VM benchmarks.

VMware Research, Remote (May 2021 - Aug 2021)
Research Intern

- Built a prototype anonymous token system on top of Concord-BFT, VMware's production Byzantine fault-tolerant distributed platform.
- Designed *Quick-Pay*, a one-round-trip low-latency payment protocol, and extended it into a lock-free sharding solution—eliminating coordination bottlenecks in distributed transaction processing.

Purdue University, West Lafayette

(Aug 2018 - Aug 2023)

Graduate Research Assistant – Department of Computer Science

- Implemented and benchmarked baseline consensus protocols (Sync HotStuff, Opt Sync, OptRand) alongside own-designed protocols across synchronous and partially synchronous network models; 10 peer-reviewed papers at CCS, NDSS, FC, and Middleware.
- Built [libnet-rs](#): a high-performance async networking library in Rust with writev-based I/O, TLS/TCP reliable transports, and a reproducible multi-VM benchmarking harness. Powers multiple BFT protocol implementations.
- First-authored *EESMR* (Middleware 2023) and implemented [libe2c-cpp](#) in C++: an energy-efficient BFT-SMR protocol that avoids quorum-certificate construction during steady-state via *voting-in-the-head*, delivering $2.8\times$ energy efficiency over Sync HotStuff in failure-free runs.
- First-authored the *Unique Chain Rule* (FC 2023): a consensus primitive where hash-chain extensions act as implicit votes, enabling **linear** communication complexity in failure-free synchronous BFT SMR (vs $O(n^2)$ in the state of the art) and $2\times$ theoretical block-production rate over Opt Sync.
- Optimized the Leto consensus hot path, delivering $6\times$ throughput improvement through a combination of lock-free data structures, buffer contiguity, and async I/O restructuring.

EDUCATION

Ph.D., Computer Science – Purdue University, West Lafayette

(2018-2023)

Advisor: Aniket Kate. Focus: Byzantine fault-tolerant distributed systems and applied cryptography.

B.Tech., Information Technology – NIT Karnataka, Surathkal

(2014-2018)

SELECTED PUBLICATIONS & PATENTS

13 peer-reviewed publications at CCS (x4), IEEE S&P, DSN, ICDCS, NDSS, Middleware, FC, and Journal of Cryptology. 3 US patents. Selected:

- **EESMR – Energy Efficient BFT-SMR for the Masses** (Middleware 2023). Leader-based SMR that avoids quorum-certificate construction during steady-state via *voting-in-the-head*; $2.8\times$ more energy-efficient than Sync HotStuff in failure-free runs.
- **The Unique Chain Rule and Its Applications** (FC 2023). Consensus primitive where hash-chain extensions serve as implicit votes; enables linear communication complexity in failure-free synchronous BFT SMR and $2\times$ block-production rate over Opt Sync.
- **Ira: Efficient Transaction Replay for Distributed Systems** (Tech report, 2026). Hint-based replay framework for primary-backup replication; $5.2\times - 23.6\times$ backup replay speedup on 24M Ethereum mainnet transactions at $\approx 11\%$ primary overhead.
- **Attacking and Improving the Tor Directory Protocol** (IEEE S\&P 2024). Identified vulnerabilities in Tor's distributed directory-consensus protocol and designed remediations; co-author.

OPEN SOURCE & SOFTWARE ARTIFACTS

- [libapollo-rs](#) (Rust) – BFT consensus testbed: reference implementations of Apollo, Artemis, Sync HotStuff, and Opt Sync with reproducible multi-VM AWS benchmarking.
- [libnet-rs](#) (Rust) – High-performance async networking library: TCP/TLS/UDP reliable transports, writev-based I/O, multi-threaded senders, stress-test harness with published baseline throughput numbers.
- [libe2c-cpp](#) (C++) – Reference implementation of EESMR (Middleware 2023): steady-state SMR without quorum certificates; $2.8\times$ more energy-efficient than Sync HotStuff.